MIL-W-38321A (USAF)
4 March 1970

SUPERSEDING
MIL-W-38321 (USAF)
1 November 1965

MILITARY SPECIFICATION

WEBBING, TEXTILE, NYLON, INTEGRAL LOCKING SLOT

1. SCOPE

1.1 Scope. This specification covers 12 types of nylon webbing.

1.2 Classification. The webbing shall be of the types specified in table I.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-W-43334	Webbing and Tape, Textile, Packaging and Packing of

STANDARDS

Federal

FED STD NO. 191	Textile Test Methods

Military

MIL-STD-105	Sampling Procedure and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 8305

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Table I. Construction and physical requirements

Requirement	Type I	Type II	Type III	Type IV	Type V	Type VI
Breaking strength 1/ (lb) (min)	1,000	2,000	3,000	4,000	5,000	6,000
Weight (oz/linear yd)	0.35±0.1	0.60±0.1	0.85±0.1	1.15±0.1	1.45±0.1	1.65±0.1
Width (inches)	5/8±1/16	5/8±1/16	1±1/16	1±1/16	1±1/16	1 1/16±1/16
Ends (full width) (min)	80	80	120	160	200	232
Picks/inch (min)	18	18	18	18	18	18
Ply (min)						
Warp	1	2	2	2	2	2
Filling	3	2	2	2	2	2
Thickness	.055±.01	.075±.01	.070±.01	.100±.01	.110±.01	.120±.01
Slot length (inches)	2+1/16-0	2+1/16-0	2+1/16-0	2+1/16-0	2+1/16-0	2+1/16-0
Solid length (inches)2/	5/8	5/8	5/8	5/8	5/8	5/8
Solid length variation 3/ (inches) (max)	1/8	1/8	1/8	1/8	1/8	1/8

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Table I. Construction and physical requirements (Continued)

Requirement	Type VII	Type VIII	Type IX	Type X	Type XI	Type XII
Breaking strength 1/ (lb) (min)	7,000	8,000	9,000	10,000	11,000	12,000
Weight (oz/linear yd)	2.10±0.15	2.40±0.15	2.95±0.15	3.35±0.15	3.55±0.15	3.75±0.15
Width (inches)	1 1/16±1/16	1 1/16±1/16	1 1/8±1/16	1 1/8±1/16	1 1/8±1/16	1 1/8±3/32
Ends (full width) (min)	112	128	150	176	192	204
Picks/inch (min)	14	14	14	13	13	13
Ply (min)						
Warp	5	5	5	5	5	5
Filling	5	5	5	5	5	5
Thickness	.170±.015	.175±.015	.185±.015	.235±.015	.250±.015	.250±.015
Slot length (inches)	2+1/16-0	2+1/16-0	2+1/16-0	2+1/16-0	2+1/16-0	2+1/16-0
Solid length 2/ (inches)	13/16	13/16	15/16	1 1/16	1 1/16	1 1/16
Solid length variation 3/ (inches) (max)	1/8	1/8	1/8	1/8	1/8	1/8

1/ The minimum breaking strength shall be the average of five breaking tests made on each sample.
2/ The average solid length is a dimensional guideline but not a physical test requirement for an individual solid length. This dimension plus the slot dimension must be controlled to provide the number of repeats within a specified length stated under paragraph 3.5.
3/ Variation between any two individual lengths within a specified length stated under paragraph 3.5.

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D.C. 20580.)

3. REQUIREMENTS

3.1 Materials

3.1.1 Yarn. The nylon yarn used in the manufacture of all types shall be a bright, high tenacity, 840/140 denier, light and heat resistant nylon prepared from hexamethylene diamine and adipic acid or its derivatives. It shall have a minimum melting point of 472°F when tested as specified in 4.5.1. The yarn shall not be bleached in any manner or process.

3.1.1.1 Ply twist. The final ply of the warp and filling yarns shall have a minimum of 2-1/2 turns per inch. The number of single yarns specified in table I shall be twisted together (plied)in one operation.

3.2 Construction and physical properties. The finished webbing shall conform to table I and 3.2.1 through 3.7.

3.2.1 Weave

3.2.1.1 In type IV warp ends weave 2 as 1 except the edges. In types V through XII, warp ends weave 4 as 1 except the edges.

3.3 Color and finish. The color of the webbing shall be natural and shall not be bleached or delustered in any manner or process. No method of treatment, finish, or conditioning shall be used which will change the frictional characteristics of the webbing as woven. (See 6.2).

3.3.1 The manufacturer shall certify that neither the yarn used nor the finished webbing has been subjected to any type of bleaching process.

3.4 Resistance to light and heat. The nylon webbing shall not lose more than 25 percent of the original breaking strength upon exposure to the light and heat tests as specified in 4.5.1.2 and 4.5.1.3.

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3.5 Repeats per radial seam length. The procuring activity shall specify a minimum and a maximum length requirement for enclosing a specified number of repeats (solid + slot, see figure 1). The number of repeats specified shall be equal to the number of slots required for the radial seam construction of the intended parachute. This requirement shall apply throughout the continuous package length or pieces of whole number multiples in a package unit as specified in paragraph 3.6. Unless otherwise specified the number of repeats per radial seam length designated by the procuring activity shall be measured as designated in 4.5.1.4.

3.6 Length of package unit (box or roll). The procuring activity shall specify the minimum continuous length of webbing for the particular parachute configuration. No piece within the package unit shall be shorter than this minimum length. Longer pieces within the package unit shall be in whole multiples of this specified length. The number of multiple lengths per package unit shall be designated by the procuring activity.

3.7 Identification tickets. Each continuous length shall have a ticket attached with not finer than 5-ply cotton string doubled to not less than 6 inches long. The tickets shall be made of paperboard not less than 0.020 inch in thickness and the color shall be manila or light in intensity to permit each reading of printed, stamped, or typed markings. The use of handwritten entries is prohibited. The ticket shall have clipped corners at the end where a reinforcing patch (with or without a metal eyelet) is firmly affixed for attaching the tying string. The ticket shall be legibly printed, using water insoluble ink, with the following information:

Specification number and title
Type
Length
Date of manufacture (month and year)
Contract number and date
Contractor's name
Repeats per radial seam length

3.7.1 Fiber identification. Each roll of webbing shall be labeled, ticketed, or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act, 3 March 1960.

3.8 Workmanship. The finished webbing shall be clean, evenly woven, and shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the levels set by the applicable acceptable quality levels.

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Table II. Classification of defects

Defect	Description	Critical	Minor
Abrasion marks	Resulting in rupture of yarns, or in nap sufficient to obscure the identity of any yarn exceeding 10 percent of width or 1 inch in length.	X	
Broken or missing end	Two or more regardless of length or a single end exceeding 6 inches in length.	X	
	Single end under 6 inches but exceeding 1/4 inch.		X
Broken or missing pick	Two or more regardless of extent.	X	
	NOTE: The filling tie-in or joining shall not be construed as a defect of any nature.		
Coarse or light filling bar	Resulting in visible difference in stiffness or thickness of webbing and extending for more than 1/4 inch in the length direction.	X	
	Resulting in visible difference in stiffness or thickness of webbing and extending for 1/4 inch or less in the length direction.		X
Twist or distortion	Webbing will not lay flat upon application of manual pressure due to twist or distortion.		X
Cut, hole, or tear	Any cut, hole, or tear.	X	
Drop-ply	Clearly visible, on more than 2 ends within same length and extending over 9 linear inches or more.	X	
	Clearly visible, on 1 or 2 ends within same length and extending over 9 linear inches or more.		X

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Table II. (Continued)

Defect	Description	Critical	Minor
Edge beaded or corded	Visible increase in edge thickness or misformed edge.		X
Edge loopy	Forming clearly visible filling loops, or edges tied loosely to body of webbing for 2 linear inches or more.	X	
Edge loose (slack)	Waviness, distortion in orientation of filling, or looseness of warp end(s) along edge.	X	
Edge cut, torn, or frayed	Clearly visible along edge.	X	
Edge tight	Visible tension along edge, or waviness, that cannot be flattened by manual pressure.	X	
Edge scalloped	Any indentation of edges above allowable tolerances.	X	
Floats or skips	Multiple, 1/2 inch or more in combined warp and filling directions or single float or skip over more than 1 inch.	X	
	Multiple, less than 1/2 inch in combined warp or filling directions or single float or skip over more than 1/2 inch but not exceeding 1 inch, if in warp, nor more than 1/4 inch of the width but not exceeding 1 inch, if in filling.		X
Hitchback crack	Clearly visible opening between adjoining picks, or warpwise tension area over part of the width resulting in visible light and heavy places.		X
Jerked-in filling, slough-off, slug	A clearly visible loop of filling pulled in at edges.		X

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Table II. (Continued)

Defect	Description	Critical	Minor
Kinks	More than 3 in any 9 linear inches.	X	
Knots	More than 1 knot in any 9 linear inches	X	
	One knot every 2 yards with untrimmed ends extending from surface of webbing.		X
Mispick, double pick	Two or more across the full width.	X	
	Single across the full width.		X
Slack end	Two or more in the same length jerked in between picks, or forming clearly visible loops on the surface.	X	
	Single jerked in between picks, or forming clearly visible loops on the surface. Single or multiple forming loops on the surface or selvage edge.		X
Slub or slug, gout	More than twice the thickness of the yarn (or ply, if plied).		X
Smash	Any smash	X	
Spot, stain, or streak	Any clearly visible dirt, rust, grease, oil spot, stain or streak.	X	
Tight end	Clearly visible up to 12 inches in length.	X	
Wrong draw	Extending for more than 9 inches	X	

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The

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Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspections. The examination and testing of the webbing shall be classified as quality conformance inspections.

4.2.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-STD-105, except as otherwise specified.

4.3 Examination of the end item

4.3.1 The lot size shall be expressed in units of the minimum continuous length repeat (see 3.6). The unit of product for this examination shall be one (1) minimum continuous length of webbing. The sample size shall be in accordance with inspection level III of MIL-STD-105. The webbing shall be examined and visual defects classified as listed in table II. The defects found shall be counted regardless of their proximity to each other except where two or more defects represent a single local condition of webbing, in which case, only the more serious defects shall be counted. A continuous defect shall be counted as one defect for each warpwise nine (9) inches or fraction thereof, in which it occurs. The acceptable quality levels (AQL) shall be 0.10 critical defect and 10 minor defects per 100 units (minimum continuous length).

4.3.2 Total package units in sample. The lot shall be unacceptable if the total of the minimum continuous length repeats in the sample is less than the total of the minimum continuous length repeats marked or package unit tickets.

4.3.3 Examination of preparation for delivery requirements. An examination shall be made to determine that packaging, packing and marking requirements of section 5 are complied with. Defects shall be scored in accordance with the list that follows. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-4 and the AQL shall be 4.0 defects per 100 units.

Examination	Defects
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.

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Examination	Defects
Weight	Net weight exceeds requirements
Workmanship	Inadequate application of components, such as: incomplete closure of case liners, container flaps, loose strapping, inadequate stapling. Bulging or distortion of containers.

4.4 Examination for compliance with Textile Fiber Products Identification Act. During the examination of individual roll for length, each roll in the sample shall be examined for conformance to the Textile Fiber Products Identification Act. Each roll not labeled in accordance with this act shall be a defect. The lot shall be unacceptable if two or more of these defects occur.

4.5 Test methods

4.5.1 Physical acceptance tests. The methods of testing shall be in accordnace with table III and as specified in 4.5.1.1 through 4.5.1.4. The physical and chemical values in section 3 apply to the average of the determinations made on a unit of product (minimum continuous length) for test purposes except for the repeats per radial seam length which shall be based on a single determination. Except for the melting point test, the unit of product for component testing shall be one cone spool or tube of yarn. The unit of product for the melting point shall be 6 inches of yarn removed from the woven webbing. The unit of product for webbing shall be three (3) radial seam length repeats. (See 3.5). The lot shall be unacceptable if one or more samples fail to meet any requirements specified. The samples shall be drawn at equally spaced production increments of the total units in each production lot.

The sample size for components and webbing shall be:

Lot size Minimum length (see 3.6)	Number of samples
Under 300	3
301-600	5
601-1000	7
1001-1800	9
1801-3200	12
3201 and over	15

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Table III. Test methods

Characteristics	Requirement reference	Fed Std 191 test method or reference
Components (yarn):		
Material	3.1.1	1/
Denier	3.1.1	1/
Tenacity	3.1.1	1/
Melting point	3.1.1	1534
Ply	Table I	Visual
Twist	3.1.1.1	4054
Webbing:		
Breaking strength	Table I	4.5.1.1
Light resistance	3.4	4.5.1.2
Heat resistance	3.4	4.5.1.3
Thickness 2/	Table I	5030
Thread count	Table I	5050
Weave	3.2.1.1	
Weight	Table I	5040
Width 3/	Table I	5020
Repeats/radial seam length	3.5	4.5.1.4

1/ Requirement accepted on basis of contractor's certificate of compliance.

2/ Except that a presser foot with a diameter of 3/8 inch, and a 6-ounce total load shall be used.

3/ The apparatus shall have adjustable caliper jaws having a jaw length and width of not less than 1.0 inch and 0.125 inches, respectively. The webbing shall be held under light hand tension. The calipers shall be held with the jaws parallel to the webbing edge and inclined, if necessary, so that the jaws are against the total thickness of the solid length portion of the webbing. The applied pressure exerted by the caliper jaws shall be sufficient to press protruding loose ends against the edge of the webbing without compression or distortion of the basic cross sectional configuration of the webbing solid.

4.5.1.1 Breaking strength of webbing. The test shall be conducted on a machine of an approved type. The pulling jaws of the machine shall separate at the rate of 4 inches per minute under no load, and the distance between jaws with the specimen in place ready for testing shall be 10 inches. The

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strength shall be reported as the average of tests on five (5) specimens of each sample.

4.5.1.2 Resistance to light. Five tests shall be made. The test specimens shall be exposed in the accelerated weathering unit as specified in method 5804 of Fed Std 191. The specimens shall be placed side by side on the rack in such a manner that only the center portion of each specimen is exposed. Corex D filters and sunshine carbons shall be used. The exposure time shall be 50 hours. The spray heads shall be shut off during the entire exposure period. At the end of the exposure period, the specimens shall be conditioned at 70° ± 2°F (21° ± 1°C) and 65 ± 2 percent relative humidity for 24 hours. They shall then be tested for breaking strength in accordance with method 4108 of Fed Std 191, and the percent of breaking strength (B.S.) loss shall be calculated as follows:

$$\frac{\text{Original B.S. - B.S. after aging}}{\text{Original B.S.}} \times 100 = \text{percent of B.S. loss}$$

4.5.1.3 Resistance to heat. Five tests shall be made. The test specimens shall be suspended in a circulating air oven at a temperature of 180° ± 3°C (356° ± 5°F) for 1 hour. After removal from the oven, the specimens shall be conditioned at 70° ± 2°F and 65 ± 2 percent relative humidity for 24 hours. They shall then be tested for breaking strength in accordance with method 4108 of Fed Std 191 and the percent of breaking strength loss shall be calculated as specified in 4.5.1.2.

4.5.1.4 Repeats per radial seam length. Types I through IV shall be measured under a tension of 40 plus or minus 2 pounds. Types V through XII shall be measured under a tension of 1 percent of the rated breaking strength plus or minus 4 pounds. Care shall be exercised in preparing length measuring equipment. If equipment such as pulleys, guides, and reels are used they shall not induce false loads or cause uneven load distribution on the webbing. Temperature and humidity conditions shall be controlled to within 55 ± 10% RH and 75° ± 5°F. The measurement shall be made within 15 seconds after application of the tension load.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. The webbing shall be packaged, packed, and marked in accordance with MIL-W-43334 except that each roll need not be over wrapped with waterproof paper. The wrapping may be in the form of a Kraft paper bag. The levels of packaging and packing shall be as specified by the procuring activity (see 6.2).

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6. NOTES

6.1 Intended use. The webbing covered by this specification is intended for use as suspension lines in ribbon parachutes.

6.2 The limited packing space for the parachute made from this webbing in the system usage requires minimum frictional characteristics for packing ease.

6.3 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification.
b. Type (see table I).
c. Minimum continuous length required (see 3.6).
d. Number of multiple lengths per roll (see 3.6).
e. Repeats per radial seam length (see 3.5).
   (1) Number -------
      a. Minimum length: _____inches
      b. Maximum length: _____inches
f. Provisions for deductions of defects (if any).
g. Applicable levels of packaging and packing (see 5.1).

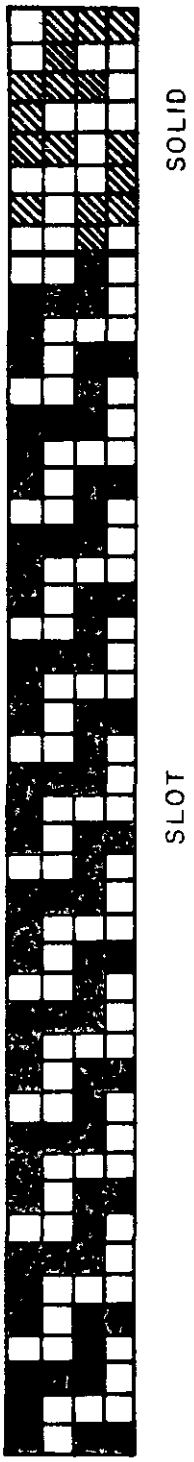
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Project 8305-F062

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NOTE: TO MAKE A SLOTTED WEBBING, TWO SHUTTLES HAVE TO BE USED, ONE FOR EACH LAYER

FIGURE 1. WEAVE DIAGRAM

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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